

LAB EXPERIMENTS

1. DDL Commands – CREATE, ALTER, DROP

AIM:

To Create, Alter, Drop, Rename, Truncate using Data Definition Language (DDL) statements.

Description:

Data Definition Language (DDL) statements are used to define the database structure or schema.

DDL Commands: Create, Alter, Drop, Rename, Truncate

CREATE:

CREATE DATABASE house_plotting;

```
Database changed
mysql> CREATE TABLE plots ( plot_id INT PRIMARY KEY, plot_no VARCHAR(10), area_sqft INT, price DECIMAL(10,2), status VARCHAR(10))
Query OK, 0 rows affected (0.03 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_house_plotting |
+-----+
| plots                     |
+-----+
1 row in set (0.01 sec)

mysql> DESC plots;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| plot_id | int | NO | PRI | NULL | |
| plot_no | varchar(10) | YES | | NULL | |
| area_sqft | int | YES | | NULL | |
| price | decimal(10,2) | YES | | NULL | |
| status | varchar(10) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.01 sec)
```

ALTER: - alters the structure of the database

ALTER TABLE plots ADD location VARCHAR(20);

```
mysql> ALTER TABLE plots ADD location VARCHAR(20);
Query OK, 0 rows affected (0.05 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> DESC plots;
```

Field	Type	Null	Key	Default	Extra
plot_id	int	NO	PRI	NULL	
plot_no	varchar(10)	YES		NULL	
area_sqft	int	YES		NULL	
price	decimal(10,2)	YES		NULL	
status	varchar(10)	YES		NULL	
location	varchar(20)	YES		NULL	

6 rows in set (0.00 sec)

DROP:

ALTER TABLE PLOTS DROP COLUMN OwnerName;

```
mysql> ALTER TABLE PLOTS DROP COLUMN OwnerName;
Query OK, 0 rows affected (0.02 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> DESC PLOTS;
```

Field	Type	Null	Key	Default	Extra
plot_id	int	NO	PRI	NULL	
plot_no	varchar(10)	YES		NULL	
area_sqft	int	YES		NULL	
price	decimal(10,2)	YES		NULL	
status	varchar(10)	YES		NULL	
location	varchar(20)	YES		NULL	

6 rows in set (0.01 sec)

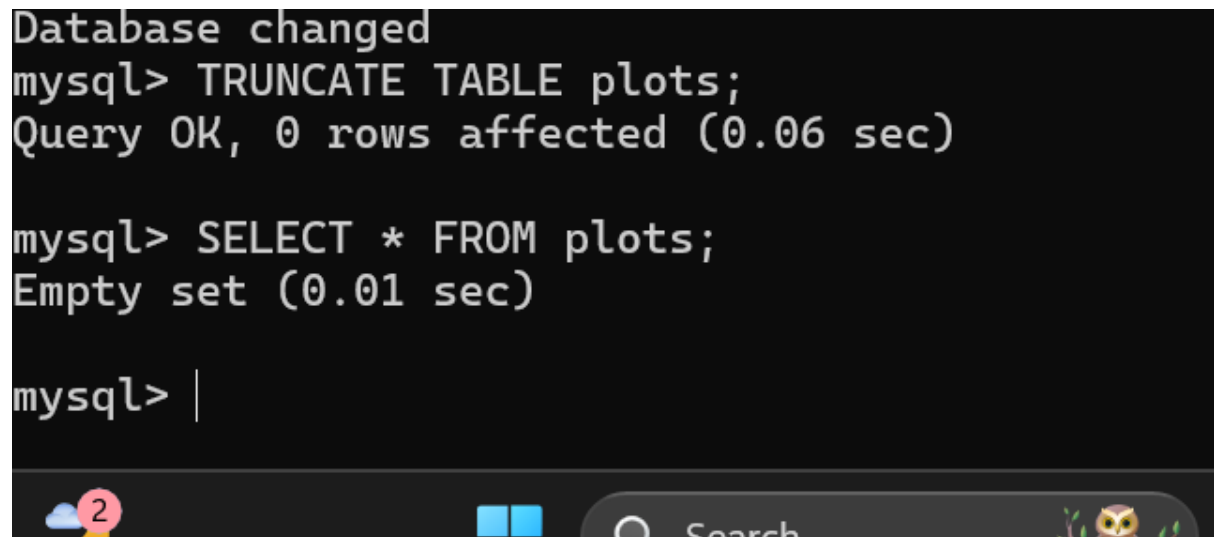
TRUNCATE:

TRUNCATE TABLE plots;

```
Database changed
mysql> TRUNCATE TABLE plots;
Query OK, 0 rows affected (0.06 sec)

mysql> SELECT * FROM plots;
Empty set (0.01 sec)

mysql> |
```



RENAME:

RENAME TABLE HOUSE_PLOTS TO PLOTS;

```
mysql> RENAME TABLE HOUSE_PLOTS TO PLOTS;
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO PLOTS VALUES ('P101',1200,1500000.00,'Available','Salem'), ('P102',1500,1800000.00,'Sold','Chennai');
ERROR 1136 (21S01): Column count doesn't match value count at row 1
mysql> DESC PLOTS;
```

Field	Type	Null	Key	Default	Extra
plot_id	int	NO	PRI	NULL	
plot_no	varchar(10)	YES		NULL	
area_sqft	int	YES		NULL	
price	decimal(10,2)	YES		NULL	
status	varchar(10)	YES		NULL	
location	varchar(20)	YES		NULL	

```
6 rows in set (0.00 sec)
```

RESULT:

Tables are created, altered and modified using DDL commands.

2. DDL Commands with Constraints – PRIMARY, FOREIGN KEY, UNIQUE, CHECK

AIM:

To add the constraints like primary key, foreign key, unique key and check using DDL commands.

Description:

PRIMARY KEY:

The PRIMARY KEY constraint uniquely identifies each record in a database table.

Primary keys must contain UNIQUE values, and cannot contain NULL values.

A table can have only one primary key, which may consist of single or multiple fields.

FOREIGN KEY:

➤ A FOREIGN KEY is a key used to link two tables together.

➤ A FOREIGN KEY is a field (or collection of fields) in one table that refers to the

PRIMARY KEY in another table.

➤ **The table containing the foreign key is called the child table, and the table containing the candidate key is called the referenced or parent table.**

UNIQUE Constraint:

➤ The UNIQUE constraint ensures that all values in a column are different.

➤ Both the UNIQUE and PRIMARY KEY constraints provide a guarantee for uniqueness for a column or set of columns.

➤ A PRIMARY KEY constraint automatically has a UNIQUE constraint.

➤ However, you can have many UNIQUE constraints per table, but only one PRIMARY KEY constraint per table.

CHECK Constraint:

➤ The CHECK constraint is used to limit the value range that can be placed in a column

➤ If you define a CHECK constraint on a single column it allows only certain values for this column.

➤ If you define a CHECK constraint on a table it can limit the values in certain columns based on values in other columns in the row.

PRIMARY:

ALTER TABLE table_name

ADD PRIMARY KEY(primary_key_column);

FOREIGN KEY:

ALTER TABLE table_name

ADD CONSTRAINT constraint_name

FOREIGN KEY foreign_key_name (columns)

REFERENCES parent_table(columns)

ON DELETE action

ON UPDATE action

UNIQUE:

CREATE TABLE table_1(

...

column_name_1 data_type,

...

UNIQUE(column_name_1)

);

CHECK

CREATE TABLE IF NOT EXISTS parts (

part_no VARCHAR(18) PRIMARY KEY,

description VARCHAR(40),

cost DECIMAL(10 , 2) NOT NULL CHECK(cost > 0), price

DECIMAL (10,2) NOT

NULL

);

Questions:

1. Add PRIMARY KEY to PLOTS

```
mysql> DESC PLOTS;
```

Field	Type	Null	Key	Default	Extra
plot_id	int	NO	PRI	NULL	
plot_no	varchar(10)	YES		NULL	
area_sqft	int	YES		NULL	
price	decimal(10,2)	YES		NULL	
status	varchar(10)	YES		NULL	
location	varchar(20)	YES		NULL	

6 rows in set (0.11 sec)

2) Alter the table name with following structure.

```
ERROR 1826 (HY000): Duplicate foreign key constraint name 'fk_plot_owner'
mysql> ALTER TABLE plots ADD FOREIGN KEY (owner_id) REFERENCES owners(owner_id);
Query OK, 0 rows affected (0.09 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> DESC plots;
```

Field	Type	Null	Key	Default	Extra
plot_id	int	NO	PRI	NULL	
plot_no	varchar(10)	YES	UNI	NULL	
area_sqft	int	YES		NULL	
price	decimal(10,2)	YES		NULL	
status	varchar(10)	YES		NULL	
location	varchar(20)	YES		NULL	
owner_id	int	YES	MUL	NULL	

7 rows in set (0.00 sec)

```
mysql> |
```



3. Add a UNIQUE constraint to plot_no.

```
mysql> SHOW CREATE TABLE plots\G
***** 1. row *****
      Table: plots
Create Table: CREATE TABLE `plots` (
  `plot_id` int NOT NULL,
  `plot_no` varchar(10) DEFAULT NULL,
  `area_sqft` int DEFAULT NULL,
  `price` decimal(10,2) DEFAULT NULL,
  `status` varchar(10) DEFAULT NULL,
  `location` varchar(20) DEFAULT NULL,
  `owner_id` int DEFAULT NULL,
  PRIMARY KEY (`plot_id`),
  UNIQUE KEY `uq_plotno` (`plot_no`),
  UNIQUE KEY `plot_no` (`plot_no`),
  KEY `owner_id` (`owner_id`),
  CONSTRAINT `fk_plot_owner` FOREIGN KEY (`owner_id`) REFERENCES `owners` (`owner_id`),
  CONSTRAINT `plots_ibfk_1` FOREIGN KEY (`owner_id`) REFERENCES `owners` (`owner_id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci
1 row in set (0.02 sec)

mysql> |
```

4. FOREIGN KEY

```
mysql> ALTER TABLE plots ADD FOREIGN KEY (owner_id) REFERENCES owners(owner_id);
Query OK, 0 rows affected (0.12 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> SHOW CREATE TABLE plots\G
***** 1. row *****
      Table: plots
Create Table: CREATE TABLE `plots` (
  `plot_id` int NOT NULL,
  `plot_no` varchar(10) DEFAULT NULL,
  `area_sqft` int DEFAULT NULL,
  `price` decimal(10,2) DEFAULT NULL,
  `status` varchar(10) DEFAULT NULL,
  `location` varchar(20) DEFAULT NULL,
  `owner_id` int DEFAULT NULL,
  PRIMARY KEY (`plot_id`),
  UNIQUE KEY `uq_plotno` (`plot_no`),
  UNIQUE KEY `plot_no` (`plot_no`),
  KEY `owner_id` (`owner_id`),
  CONSTRAINT `fk_plot_owner` FOREIGN KEY (`owner_id`) REFERENCES `owners` (`owner_id`),
  CONSTRAINT `plots_ibfk_1` FOREIGN KEY (`owner_id`) REFERENCES `owners` (`owner_id`),
  CONSTRAINT `plots_ibfk_2` FOREIGN KEY (`owner_id`) REFERENCES `owners` (`owner_id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci
1 row in set (0.00 sec)

mysql> |
```


3.DML Commands – INSERT, SELECT

Aim:

DML Commands – INSERT, SELECT

To perform Data Manipulation Language (DML) Commands such as INSERT, SELECT in the table.

Description:

Data Manipulation Language (DML) statements are used for managing data within schema objects. DML

Commands: Insert, Select

- INSERT - insert data into a table
- SELECT - retrieve data from the a database

INSERT:

INSERT INTO table_name

VALUES (value1, value2, value3,...);

(or)

INSERT INTO table_name (column1, column2, column3,...)

VALUES (value1, value2, value3,...);

SELECT:

```
SELECT column_name(s)
FROM table_name;
```

Questions:

1. Populate all the five tables with your own data.
2. View all the records from the five tables.

OUTPUT:

Table 1: PLOTS

INSERT INTO plots VALUES

```
(1,'P101',1200,1500000.00,'Available','Salem',1),
(2,'P102',1500,1800000.00,'Sold','Chennai',2),
(3,'P103',1000,1300000.00,'Available','Erode',1);
```

```
2 rows in set (0.02 sec)

mysql> INSERT INTO plots (plot_id, plot_no, area_sqft, price, status, location, owner_id) VALUES (1,'P101',1200,1500000.00,'Available','Salem',1), (2,'P102',1500,1800000.00,'Sold','Chennai',2), (3,'P103',1000,1300000.00,'Available','Erode',1);
Query OK, 3 rows affected (0.02 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM plots;
+----+-----+-----+-----+-----+-----+-----+
| plot_id | plot_no | area_sqft | price | status | location | owner_id |
+----+-----+-----+-----+-----+-----+-----+
| 1 | P101 | 1200 | 1500000.00 | Available | Salem | 1 |
| 2 | P102 | 1500 | 1800000.00 | Sold | Chennai | 2 |
| 3 | P103 | 1000 | 1300000.00 | Available | Erode | 1 |
+----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> |
```

Question 2: View all the records from the five tables.

SELECT * FROM owners; SELECT * FROM plots;

```
mysql> SELECT * FROM payments;
+-----+-----+-----+-----+
| payment_id | plot_id | amount | payment_date |
+-----+-----+-----+-----+
| 101 | 1 | 500000.00 | 2026-08-01 |
| 102 | 2 | 1000000.00 | 2026-08-02 |
| 103 | 3 | 300000.00 | 2026-08-03 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
mysql> CREATE TABLE registrations (
  -> reg_id INT,
  -> owner_id INT,
  -> plot_id INT,
  -> reg_date DATE
  -> );
Query OK, 0 rows affected (0.02 sec)

mysql> INSERT INTO registrations VALUES (201,1,1,'2026-08-01'), (202,2,2,'2026-08-02'), (203,1,3,'2026-08-03'); SELECT * FROM registrations;
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

+-----+-----+-----+-----+
| reg_id | owner_id | plot_id | reg_date |
+-----+-----+-----+-----+
| 201 | 1 | 1 | 2026-08-01 |
| 202 | 2 | 2 | 2026-08-02 |
| 203 | 1 | 3 | 2026-08-03 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```